PHENet: Toward Robust Fog-Resilient Change Detection in Remote Sensing Imagery With a Physics-Guided Height-Enhanced Network

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***Abstract*—The change detection from remote sensing imagery faces significant challenges under foggy conditions, where fog and haze severely degrade image quality and compromise detection accuracy. Current methods for foggy conditions rely on the generic image enhancement without considering the physics-based correlation between fog distribution and terrain elevation. This article presents PHENet, a physics-guided height-enhanced network that leverages the intrinsic correlation between atmospheric scattering and terrain elevation for robust change detection under foggy conditions. The proposed approach introduces three synergistic components: 1) height-aware mixture fusion (HAMF) module that leverages frequency-domain decomposition for elevation-conditioned spatial feature enhancement, selectively suppressing fog-related artifacts while preserving structural details; 2) height-enhanced difference (HED)-augmented difference fusion module that incorporates height-derived change indicators invariant to atmospheric variations, significantly reducing false positives from weather-induced spectral changes; and 3) physics-guided adversarial refinement module (PGARM) that enforces consistency with atmospheric scattering principles during training without requiring paired clear-foggy data. Comprehensive experiments on foggy variants of WHU-CD and LEVIR-CD datasets demonstrate that PHENet achieves superior performance across all fog densities. Furthermore, the proposed method maintains a compact architecture with 5.24M parameters, demonstrating that physics-informed design can deliver robust performance without excessive model complexity. This code can be available at: https://github.com/sssy3/PHENet**

***Index Terms*—Change detection, foggy weather conditions, height estimation, physics-guided learning.**

I. Introduction

CHANGE detection from multitemporal remote sensing imagery has become an indispensable tool for urban monitoring, disaster assessment, and sustainable development planning. With the exponential growth of satellite data availability and computational capabilities, deep learning methods have emerged as the dominant paradigm for automated change detection.

Contemporary deep learning architectures for change detection employ diverse strategies to model temporal relationships in remote sensing data. Foundational approaches based on Siamese convolutional neural networks [1] establish dual-stream processing pipelines that extract features from bitemporal images independently before fusion. Subsequent developments integrate attention mechanisms to enhance discriminative capabilities, with DSAMNet [2] achieving significant improvements through deep supervision and metric learning. Recent advances have also explored frequency-domain analysis, with approaches like CDC2F [3] leveraging the discrete cosine transform to capture both spatial and frequency information for enhanced change detection performance. The introduction of transformer architectures marks a paradigm shift in change detection, with BIT [4] and ChangeFormer [5] demonstrating superior performance through global context modeling and self-attention mechanisms. Most recently, state-space models like ChangeMamba [6] offer efficient alternatives with linear complexity while maintaining the competitive accuracy.

However, foggy conditions pose critical challenges for change detection. Fog and haze severely degrade image quality through atmospheric scattering, resulting in reduced contrast, color distortion, and loss of structural details essential for accurate change identification [7]. While recent methods [8] address adverse conditions through learnable image filters, they treat fog as generic image degradation without exploiting the physical relationship between atmospheric scattering and terrain elevation, limiting their ability to distinguish fog-induced artifacts from genuine structural changes.

The relationship between atmospheric scattering and terrain elevation presents an unexploited opportunity for robust change detection. The intensity of atmospheric scattering varies predictably with elevation due to altitude-dependent pressure and particle density [9]. Recent advances in monocular height estimation [10] enable practical extraction of elevation information from remote sensing imagery, while multimodal fusion strategies have shown promise in related tasks [11]. This physics-based understanding motivates the integration of height information as a complementary information for the detection of robust fog changes.

To address these limitations, this article proposes PHENet, a physics-guided height-enhanced framework that exploits the intrinsic correlation between terrain elevation and fog distribution. The framework integrates three synergistic strategies: frequency-domain processing to exploit fog's spectral characteristics, height-aware feature fusion to leverage

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elevation-invariant features, and physics-based adversarial learning to embed atmospheric constraints without requiring paired clear-foggy training data. By combining these complementary approaches, PHENet addresses the fundamental ambiguity between atmospheric degradation and actual scene changes, enabling robust detection under varying atmospheric conditions. Extensive experiments demonstrate that PHENet achieves performance superior to that of the state of the art. The main contributions of this article are as follows.

1) This work addresses the remote sensing change detection (RSCD) under foggy conditions by exploiting the physical correlation between terrain elevation and atmospheric scattering intensity. To the best of our knowledge, this is the first approach to leverage this relationship for the detection of robust fog changes.
2) The height-aware mixture fusion (HAMF) module employs frequency-domain decomposition with elevation-guided enhancement to suppress fog artifacts while preserving structural details.
3) The HED-augmented difference fusion module (HED) integrates spectral and height-based change indicators to reduce false positives from atmospheric variations.
4) The physics-guided adversarial refinement module (PGARM) embeds atmospheric scattering constraints during training without requiring paired clear-foggy data.

II. Related Works

A. Deep Learning-Based RSCD

Deep learning has become the dominant paradigm for RSCD. Early CNN-based methods established fundamental architectures and benchmarks. STANet [12] introduced spatial–temporal attention and the LEVIR-CD dataset, while subsequent studies improved multiscale representation and optimization through deep supervision [13], dense connectivity [14], and explicit modeling of changed and unchanged regions [15].

To overcome the locality of convolutions, attention mechanisms were incorporated for long-range dependency modeling. Representative works include DASNet [16], H-SALENet [17], and DSAMNet [2]. Recent studies further emphasized change-aware representation learning. HANet [18] addressed severe class imbalance via foreground-aware sampling and lightweight hierarchical attention, while HSANet [19] employed hybrid self-attention and cross-attention to enhance cross-temporal feature interaction.

Transformer-based architectures enabled global contextual modeling in RSCD. BIT [4] and ChangeFormer [5] demonstrated the effectiveness of Transformer-based bitemporal interaction. Beyond generic designs, MDGF-CD [20] embedded temporal difference information directly into the attention mechanism to strengthen change sensitivity. In parallel, ChangeCLIP [21] explored vision–language pretraining for semantic change understanding, and ChangeMamba [6] leveraged state-space models for efficient long-range modeling.

Reducing the reliance on dense pixel-level annotations has become an important research direction. SWCD [22] introduced similarity-aware weak supervision using region-level labels to guide pixel-level change detection, whereas LGTC [23] proposed a local–global triconsistency learning framework that enforces consistency at pixel, region, and image levels under limited supervision.

Recently, foundation models have shown potential for generalized remote sensing analysis. HyperFree [24] proposed a channel-adaptive, tuning-free foundation model for hyperspectral imagery, indicating a promising direction for label-efficient change detection. Nevertheless, most existing RSCD methods remain appearance-driven and vulnerable to atmospheric degradation. To the best of our knowledge, IA-CDNet [8] is currently the only method explicitly designed for change detection under adverse atmospheric conditions such as fog or haze. While IA-CDNet adopts adaptive image filtering strategies, it does not explicitly model the physical imaging process. In contrast, the proposed PHENet incorporates atmospheric scattering physics and terrain elevation priors to enable robust change detection under foggy conditions.

B. Height Estimation

Height estimation from remote sensing imagery derives elevation information to provide crucial 3-D structural data complementing traditional spectral information. These elevation products, typically represented as digital surface models or heightmaps, are increasingly vital for comprehensive scene understanding in remote sensing applications.

Deep learning has revolutionized monocular height estimation, eliminating dependencies on stereo imagery or LiDAR data. Early CNN-based approaches like [25] demonstrated effective learning of complex appearance elevation mappings through implicit geometric cues. Subsequent advances addressed scale ambiguity through classification regression paradigms, with HTC-DC Net [26] introducing hierarchical context modeling for long-tailed height distributions in urban environments. Recent transformer architectures mark a paradigm shift: MFTSC [27] combines Swin Transformers with CNN-based spatial features through cross-token attention fusion, while 3DCentripetalNet [28] introduces 3-D centripetal shift representations unifying planar and vertical structures, achieving superior performance over traditional pixelwise regression.

The height integration significantly enhances various remote sensing tasks by resolving spectral ambiguities through geometric constraints. DSMFNet [29] demonstrates consistent improvements in road-rooftop distinction and vegetation differentiation through lightweight DSM fusion branches. AFNet [30] achieves enhanced boundary delineation in complex urban scenes via adaptive multipath attention fusion of height and spectral features. These advances establish height information as an essential complementary information for addressing challenging remote sensing scenarios.

C. Atmospheric Degradation Mitigation in Vision Tasks

Processing foggy and hazy images remains a fundamental challenge in vision tasks, particularly for RSCD, where atmospheric conditions significantly impact detection

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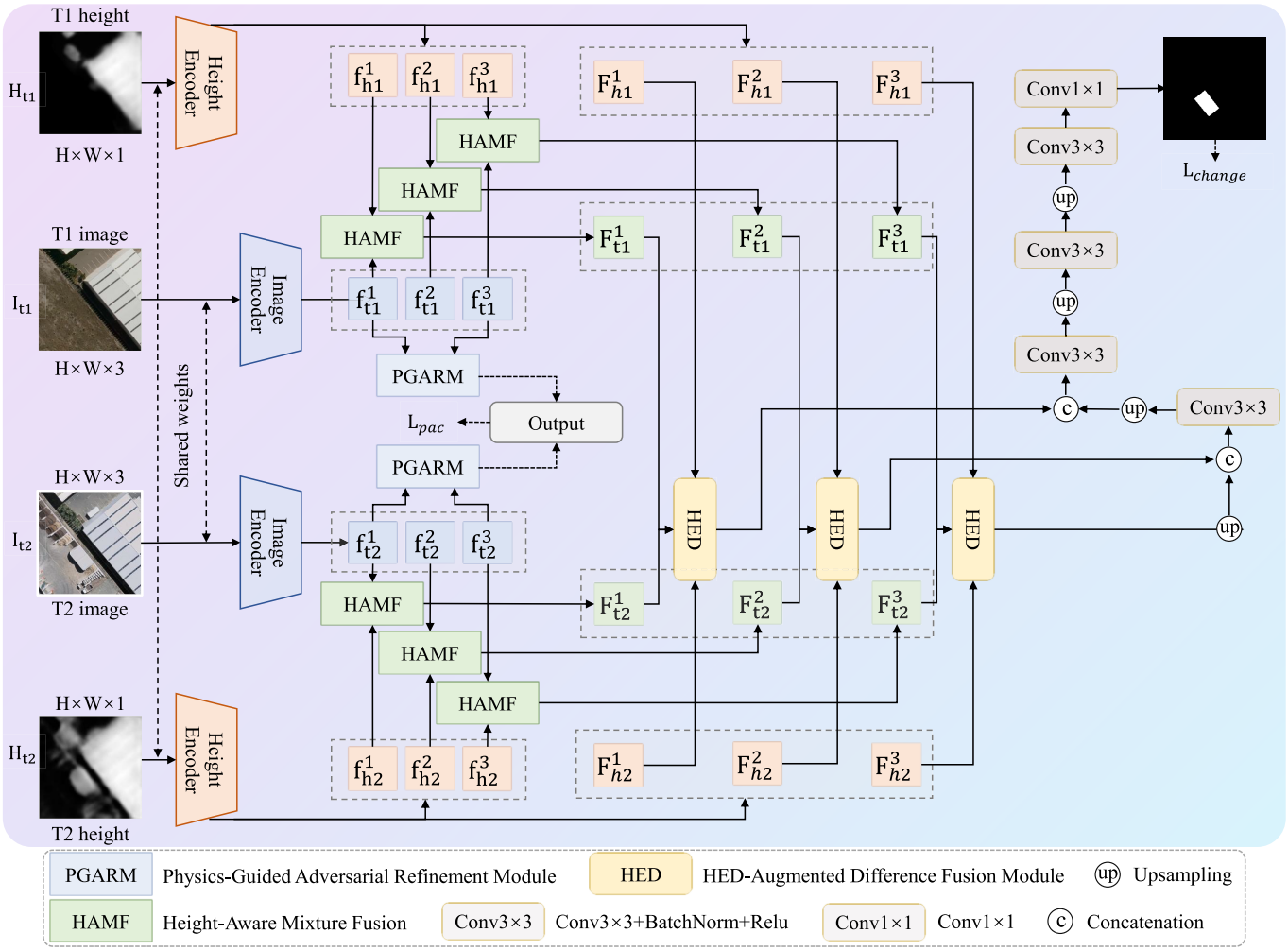


Fig. 1. Overview of the proposed PHENet framework.

accuracy. The atmospheric scattering model [31] provides the theoretical foundation for understanding fog formation, which early methods exploited through physics-based priors like the dark channel [32].

Deep learning approaches revolutionized this field by shifting from physical modeling to data-driven solutions. While initial networks focused on restoration quality via multiscale processing [33], selective feature emphasis [34], and atmospheric constraint integration [35], recent advances like DEA-Net [36] and UCL-Dehaze [37] have achieved superior restoration through reparameterization and contrastive learning.

However, optimal image restoration does not guarantee optimal task performance. DSNet [38] demonstrated that joint optimization of dehazing and detection significantly outperforms sequential approaches, while BAD-Net [39] addressed variable haze densities through adaptive fusion. For change detection, the temporal consistency is crucial as varying atmospheric conditions between acquisitions can create pseudo-changes. Current methods [8] uniformly process images without considering elevation-dependent atmospheric variations, motivating our physics-guided approach that integrates height information to address these limitations.

III. Methodology

A. Overall Architecture

The proposed PHENet framework employs a dual encoder architecture to jointly process bitemporal remote sensing images (I_{t1}, I_{t2}) and their corresponding heightmap (H_{t1}, H_{t2}), as illustrated in Fig. 1. The image encoder, implemented as a lightweight MobileNetV2 backbone, extracts multiscale features f_{tj}^{i} where $j \in \{1, 2\}$ denotes the temporal phase and $i \in \{1, 2, 3\}$ represents the feature hierarchy from shallow to deep. In parallel, the height encoder employs a dual-branch design tailored for single-channel elevation data, with each branch consisting of a sequential module containing two convolutional layers with ReLU activation followed by adaptive average pooling. Specifically, for each feature level i, both branches transform the input heightmap through progressive channel expansion and spatial alignment to match the dimensions of the corresponding RGB features. The first branch generates features f_{hj}^{i} for fusion enhancement within the HAMF module, while the second produces F_{hj}^{i} for subsequent extraction of difference features in the HED module. This dual-branch design ensures that height information can be

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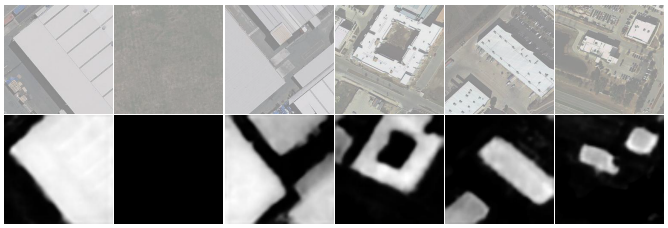


Fig. 2. Foggy images and their estimated heightmaps. Higher elevations appear brighter.

optimally utilized for different purposes without interference between the two processing pathways.

Since existing change detection datasets lack elevation annotations, we employ a pretrained Swin-UNet model for height estimation. This model was trained on a custom dataset built from aerial orthoimagery and corresponding LiDAR-derived normalized digital surface models (nDSMs) from the Valladolid region, accessed through the open-data portal of Spanish National Center for Geographic Information. The nDSMs, processed using QGIS from contemporaneous LiDAR data, provide ground-truth elevation information. This domain-specific training ensures that the height estimation model produces reliable elevation features for our target remote sensing scenarios, as demonstrated in Fig. 2.

The extracted features are then processed through three synergistic modules. The HAMF enhances RGB features using height guidance to suppress fog artifacts while preserving structural details crucial for change identification. Subsequently, the HED generates robust change representations by adaptively combining enhanced RGB features with height-based change indicators. During training, the PGARM operates on shallow and deep features to enforce atmospheric scattering constraints, embedding fog-handling capabilities into the network without affecting inference efficiency. The decoder network progressively upsamples and fuses the multi-scale difference features through convolution layers to generate the final binary change map.

B. Height-Aware Mixture Fusion

The HAMF module leverages the inherent correlation between atmospheric scattering and terrain elevation to perform spatially adaptive feature enhancement. As illustrated in Fig. 3, the module processes multiscale RGB features f_{tj}^i and height features f_{hj}^i extracted from the backbone network.

To exploit the distinct spectral characteristics of fog, we employ frequency-domain analysis with decomposition in the frequency domain followed by elevation-guided enhancement in the spatial domain. The atmospheric scattering predominantly affects low-frequency components, manifesting as uniform haze that reduces global contrast, while high-frequency details such as edges and textures are attenuated but not eliminated. This frequency-dependent degradation pattern motivates our dual-pathway design that processes low and high-frequency components separately with elevation-guided attention mechanisms. Specifically, both RGB and height features are first decomposed into frequency bands through

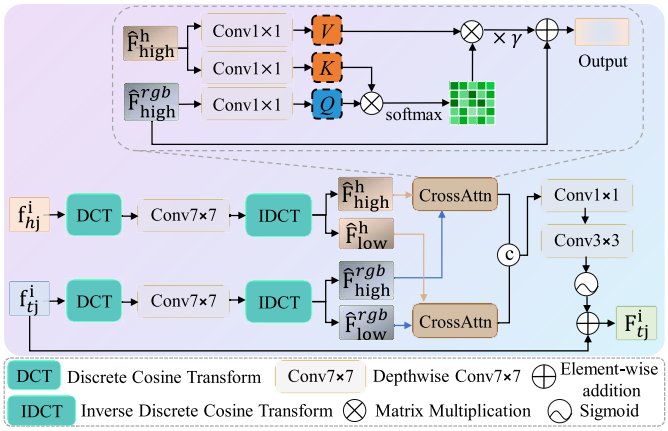


Fig. 3. Illustration of the HAMF module.

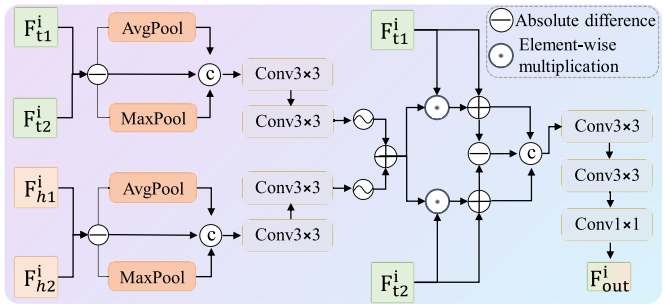


Fig. 4. Illustration of the HED module.

learnable DCT operations, followed by IDCT to reconstruct the spatial features

$$\hat{F}_{\text{high}}^{\text{rgb}}, \hat{F}_{\text{low}}^{\text{rgb}} = \text{IDCT}\left(\text{Conv}_{7\times7}\left(\text{DCT}\left(f_{tj}^i\right)\right)\right) \tag{1a}$$

$$\hat{F}_{\text{high}}^{h}, \hat{F}_{\text{low}}^{h} = \text{IDCT}\left(\text{Conv}_{7\times7}\left(\text{DCT}\left(f_{hj}^i\right)\right)\right). \tag{1b}$$

The reconstructed features are then enhanced through cross-attention mechanisms, where the attention weights learned from height features guide the network to suppress fog-related low-frequency components while enhancing structural details in the high-frequency domain. Finally, the enhanced features from both frequency bands are adaptively fused

$$F_{tj}^i = \sigma\left(\text{Conv}_{3\times3}\left(\text{Conv}_{1\times1}\left(\left[\text{CrossAttn}\left(\widehat{F}_{\text{high}}^{\text{rgb}}, \widehat{F}_{\text{high}}^{h}\right), \text{CrossAttn}\left(\widehat{F}_{\text{low}}^{\text{rgb}}, \widehat{F}_{\text{low}}^{h}\right)\right]\right)\right)\right) + f_{tj}^i \tag{2}$$

where σ denotes the sigmoid activation, $[\cdot]$ represents the concatenation, and CrossAttn denotes the cross-attention mechanism.

C. HED-Augmented Difference Fusion

The HED module addresses a critical challenge in foggy change detection: distinguishing genuine structural changes from atmospheric-induced variations. As illustrated in Fig. 4, the module processes refined RGB features F_{t1}^i and F_{t2}^i from both temporal phases, along with corresponding height features F_{h1}^i and F_{h2}^i. The module first computes temporal difference features in both feature representations

$$\Delta F_t = \left|F_{t1}^i - F_{t2}^i\right| \tag{3a}$$

$$\Delta F_h = \left|F_{h1}^i - F_{h2}^i\right|. \tag{3b}$$

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These different features capture distinct change patterns: ΔF_t identifies spectral variations that may include both genuine changes and atmospheric artifacts, while ΔF_h detects structural modifications that remain invariant to illumination and weather conditions. This complementary information forms the basis for robust change detection under foggy conditions. The difference features undergo parallel processing through dual-pathway attention generation

$$w_{\text{rgb}} = \sigma(\text{Conv}_{3\times3}(\text{Conv}_{3\times3}([\Delta F_t, \text{AvgPool}(\Delta F_t), \text{MaxPool}(\Delta F_t)]))) \tag{4a}$$

$$w_h = \sigma(\text{Conv}_{3\times3}(\text{Conv}_{3\times3}([\Delta F_h, \text{AvgPool}(\Delta F_h), \text{MaxPool}(\Delta F_h)]))). \tag{4b}$$

The integration of height information through w_h provides robustness against illumination variations and shadow effects that commonly cause false positives in RGB-only methods. The combined weight ($w = w_{\text{rgb}} + w_h$) forms a unified attention mechanism that highlights regions exhibiting both spectral and geometric changes

$$\tilde{F}^i_{t_1} = F^i_{t_1} \cdot (1 + w) \tag{5a}$$

$$\tilde{F}^i_{t_2} = F^i_{t_2} \cdot (1 + w). \tag{5b}$$

The final output fuses the enhanced features with the original bitemporal representations, enabling the network to capture both fine-grained changes and broader context for improved detection accuracy

$$F^i_{\text{out}} = \text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(\left[\tilde{F}^i_{t1}, \tilde{F}^i_{t2}, \left|\tilde{F}^i_{t1} - \tilde{F}^i_{t2}\right|\right]\right)\right)\right). \tag{6}$$

D. Physics-Guided Adversarial Refinement Module

The PGARM leverages atmospheric scattering physics and adversarial training to enhance feature representations without requiring clean reference images. The module incorporates a physics-guided auxiliary branch that models atmospheric degradation and reconstructs fog-affected imagery, enabling the network to distinguish weather-induced artifacts from genuine structural changes. As shown in Fig. 5, the module processes features f^1_{tj} and f^3_{tj} through a physics-based auxiliary branch that leverages the atmospheric scattering model

$$I = J \cdot t + A \cdot (1 - t) \tag{7}$$

where I represents the observed foggy image, J denotes the underlying clear scene, A is the atmospheric light, and $t = e^{-\beta d}$ represents the transmission map with β controlling the scattering coefficient and d representing the scene depth. The multiscale features are first fused via channel attention (SEBlock) and convolutional operations to estimate these physical parameters and reconstruct the foggy image

$$F^{\text{fused}}_{tj} = \text{Conv}_{1\times1}\left(\text{up}\left(\text{SEBlock}\left(\text{Conv}_{3\times3}\left(\left[f^1_{tj}, \text{Conv}_{1\times1}\left(\text{up}\left(f^3_{tj}\right)\right)\right]\right)\right)\right)\right) \tag{8}$$

$$t_j = \sigma\left(\text{Conv}_{3\times3}\left(F^{\text{fused}}_{tj}\right)\right) \tag{9}$$

$$A_j = \text{AvgPool}\left(F^{\text{fused}}_{tj}\right) \tag{10}$$

$$\hat{I}_{tj} = \text{Conv}_{1\times1}\left(F^{\text{fused}}_{tj}\right) \cdot t_j + A_j \cdot \left(1 - t_j\right). \tag{11}$$

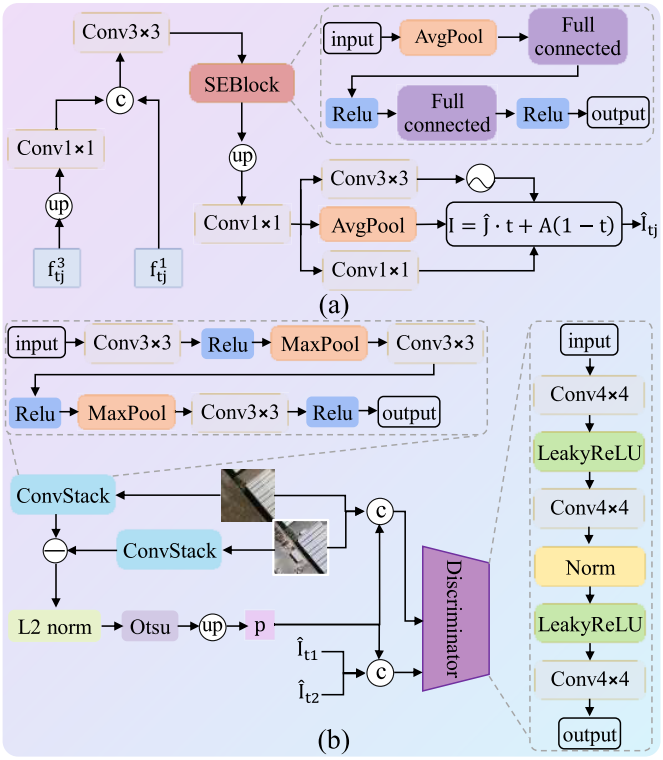


Fig. 5. (a) Illustration of the PGARM. (b) Illustration of the adversarial framework.

The reconstructed foggy image $\hat{I}_{tj}$ serves as the generator's output in the adversarial framework, preserving structural details while maintaining realistic fog characteristics. Subsequently, the potential clear image J_{tj}, which is used exclusively for dark channel loss computation, is recovered through the inverse atmospheric scattering model

$$J_{tj} = \frac{\hat{I}_{tj} - A_j \cdot (1 - t_j)}{t_j + \epsilon} \tag{12}$$

where ϵ is a small constant for numerical stability. The physical constraints are formulated through two loss functions

$$\mathcal{L}^{(j)}_{tv} = \sum_{m,n} |t_j(m+1,n) - t_j(m,n)| + \sum_{m,n} |t_j(m,n+1) - t_j(m,n)| \tag{13}$$

$$\mathcal{L}^{(j)}_{\text{dark}} = \mathbb{E}\left[\min_c J^c_{tj}(x)\right]^2 \tag{14}$$

where $\mathbb{E}[\cdot]$ denotes the expectation operator. The total variation loss enforces the spatial smoothness of t_j, reflecting that the density of fog varies gradually in space. The dark channel loss ensures that J_{tj} satisfies the dark channel prior, a fundamental characteristic of natural haze-free images.

The adversarial framework employs pseudo-labels p to guide the discriminator toward change-relevant regions. The discriminator loss is exclusively employed to optimize discriminator parameters through independent gradient updates, while the generator loss is integrated into the aggregated

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TABLE I
OVERVIEW INFORMATION ABOUT THE TWO PUBLIC CD DATASETS

Datasets	Image pairs	Original image size	Space resolution	Train/Val/Test	Image type
WHU-CD	1	32207×15354	0.2 m/pixel	6096/760/760	Aerial images
LEVIR-CD	637	1024×1024	0.5 m/pixel	7,120/1,024/2,048	Satellite images

physics-guided adversarial consistency (PAC) loss to jointly update both the primary network and PGARM

$$\mathcal{L}_{\text{adv}}^{G} = -\mathbb{E}\left[\log \mathcal{D}\left(\left[\hat{I}_{t1}, \hat{I}_{t2}, p\right]\right)\right] \tag{15}$$

$$\mathcal{L}_{\text{pac}} = \lambda_1 \left(\mathcal{L}_{tv}^{(1)} + \mathcal{L}_{tv}^{(2)}\right) + \lambda_2 \left(\mathcal{L}_{\text{dark}}^{(1)} + \mathcal{L}_{\text{dark}}^{(2)}\right) + \lambda_3 \mathcal{L}_{\text{adv}}^{G} \tag{16}$$

where $\mathcal{L}_{\text{adv}}^{G}$ encourages the realistic foggy image reconstruction and λ_i is the weighting coefficients that balance different loss components. The physics-guided constraints ensure that refined features maintain physical plausibility, while adversarial training enhances fine-grained details crucial for accurate change boundary delineation. The parameters are set as follows: $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.5$.

E. Loss Function

The overall training objective of our framework integrates the primary change detection loss with the PAC loss $\mathcal{L}_{\text{pac}}$ to ensure both accurate change identification and resilience to fog. The change detection loss $\mathcal{L}_{\text{change}}$ employs a hybrid of binary cross entropy (BCE) $\mathcal{L}_{\text{BCE}}$ and Dice loss $\mathcal{L}_{\text{Dice}}$ to mitigate class imbalance. The total objective is formulated as follows:

$$\mathcal{L}_{\text{change}} = 1 \times \mathcal{L}_{\text{BCE}} + 1 \times \mathcal{L}_{\text{Dice}} \tag{17}$$

$$\mathcal{L}_{\text{total}} = 1 \times \mathcal{L}_{\text{change}} + 1 \times \mathcal{L}_{\text{pac}}. \tag{18}$$

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: The evaluation of change detection methods under foggy conditions requires specialized datasets that capture the degradation effects of atmospheric scattering. However, no publicly available change detection datasets with natural fog variations currently exist. To address this limitation, foggy imagery is synthesized from two widely used change detection benchmarks: WHU-CD [40] and LEVIR-CD [12]. The details of two datasets are shown in Table I.

To simulate realistic foggy conditions, the atmospheric scattering model is applied as described in (7). Three fog density levels are generated by varying the scattering coefficient $\beta \in \{0.4, 0.6, 2\}$ while maintaining constant atmospheric light $A = 0.6$, corresponding to light, medium, and heavy fog conditions, respectively. This variation enables a comprehensive evaluation of method robustness across different visibility conditions, yielding six synthetic foggy datasets.

2) Implementation Details: The proposed framework is implemented using PyTorch and trained on an NVIDIA Titan V GPU. The network is optimized with Adam at an initial learning rate of 1×10^{-4} for 300 epochs with a batch size of 8. To ensure fair comparison, all methods employ identical data augmentation strategies, including random flipping, rotation, scale variation, cropping, Gaussian blur, and color jittering.

3) Evaluation Metrics: The performance of change detection methods is evaluated using four standard binary classification metrics: precision (P), recall (R), $F1$-score ($F1$), and intersection over union (IoU). They are defined as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{22}$$

where TP, FP, and FN denote the true positives, false positives, and false negatives, respectively. In all subsequent results tables, all metric values are expressed as percentages, with the best value for each metric typed in bold and the second-best underlined.

B. Comparison With State-of-the-Art Methods

To rigorously evaluate the proposed method's robustness to foggy conditions, comprehensive comparisons are conducted with eight state-of-the-art change detection methods. The evaluated methods include the following: MSCANet [41], ACABFNet [42], IA-CDNet [8], GAS-Net [43], AMTNet [44], DGMA2-Net [45], STFF-GA [46], and CASP [47].

1) Quantitative Analysis: The quantitative evaluation reveals PHENet's robust performance across all fog conditions on WHU-CD and LEVIR-CD datasets, as detailed in Tables II and III. Across both datasets, PHENet achieves the highest IoU scores in most scenarios while maintaining remarkable stability. On the WHU-CD dataset, PHENet attains 88.36%, 87.92%, and 87.58% IoU for light, medium, and heavy fog conditions, respectively, representing improvements of 0.35%, 1.14%, and 1.98% over the second-best methods. The performance degradation from light to heavy fog is notably minimal with only a 0.78% IoU drop, indicating robust fog-handling capabilities.

The LEVIR-CD results reveal interesting patterns in method robustness. While CASP performs competitively in light fog conditions with 83.18% IoU, its performance degrades more severely under heavy fog to 80.10% IoU. In contrast, PHENet maintains more stable performance with 83.25%, 83.03%, and 82.13% IoU across fog densities, exhibiting only 1.12% degradation. This stability is particularly valuable

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TABLE II
COMPARISON WITH OTHER SOTA MODELS ON THE WHU-CD DATASET UNDER VARYING FOG DENSITIES

Models	Params (M)↓	light-foggy				medium-foggy				heavy-foggy			
		P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑
IA-CDNet	12.60	91.21	61.39	73.39	57.96	81.08	73.69	77.21	62.88	75.63	80.18	77.84	63.72
ACABFNet	101.06	88.41	86.27	87.32	77.50	89.18	84.42	86.73	76.58	72.54	48.13	57.86	40.71
AMTNet	16.44	92.64	88.61	90.58	82.78	91.31	89.38	90.34	82.37	91.71	86.69	89.13	80.39
MSCANet	16.42	94.19	88.12	91.06	83.58	92.73	88.72	90.68	82.95	91.25	86.98	89.07	80.29
GASNet	23.52	92.12	**92.86**	92.49	86.02	82.09	**92.53**	92.31	85.72	90.50	91.84	91.17	83.78
DGMA2	37.10	93.86	91.69	92.76	86.50	93.90	91.09	92.47	86.00	94.06	90.49	92.24	85.60
CASP	**4.57**	94.97	91.67	93.29	87.43	**95.55**	90.43	92.92	86.78	87.96	79.50	83.52	71.70
STFFGA	12.28	**95.82**	91.52	93.62	88.01	94.07	91.68	92.86	86.67	94.60	89.60	92.03	85.24
PHENet	5.24	95.02	92.65	**93.82**	**88.36**	95.41	91.81	**93.57**	**87.92**	**94.97**	**91.85**	**93.38**	**87.58**

TABLE III
COMPARISON WITH OTHER SOTA MODELS ON THE LEVIR-CD DATASET UNDER VARYING FOG DENSITIES

Models	Params (M)↓	light-foggy				medium-foggy				heavy-foggy			
		P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑
IA-CDNet	12.60	85.55	86.96	86.25	75.82	88.35	85.23	86.76	76.62	87.29	83.91	85.56	74.77
ACABFNet	101.06	87.47	87.85	87.66	78.03	87.07	88.21	87.64	77.99	85.46	71.62	77.93	63.84
AMTNet	16.44	86.87	87.71	87.29	77.44	88.90	85.45	87.14	77.22	87.29	84.53	85.89	75.26
MSCANet	16.42	88.86	85.43	87.11	77.17	88.50	85.60	87.03	77.03	87.60	83.52	85.51	74.69
GASNet	23.52	89.13	**92.00**	90.54	82.72	87.54	**93.31**	90.33	82.37	88.23	**91.36**	89.77	81.44
DGMA2	37.10	91.12	89.41	90.26	82.25	91.23	88.48	89.84	81.55	89.98	88.31	89.14	80.40
CASP	**4.57**	**92.63**	89.07	90.81	83.18	92.16	88.77	90.43	82.53	89.68	88.23	88.95	80.10
STFFGA	12.28	92.16	89.32	90.72	83.01	92.12	88.82	90.44	82.55	91.00	88.73	89.85	81.57
PHENet	5.24	91.88	89.86	**90.86**	**83.25**	**93.00**	88.56	**90.73**	**83.03**	**92.16**	88.29	**90.19**	**82.13**

TABLE IV
COMPARISON WITH OTHER SOTA MODELS ON WHU-CD AND LEVIR-CD UNDER CLEAR CONDITIONS

Models	WHU-CD				LEVIR-CD			
	P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑
IA-CDNet	67.31	64.11	65.67	48.89	85.52	80.56	82.97	70.89
ACABFNet	89.45	85.84	87.61	77.95	89.32	86.35	87.81	78.27
AMTNet	92.97	88.42	90.64	82.88	89.08	85.99	87.50	77.78
MSCANet	94.43	88.22	91.22	83.86	89.11	86.46	87.77	78.21
GASNet	93.40	92.74	93.07	87.04	92.14	89.17	90.63	82.87
DGMA2	94.50	91.35	92.90	86.73	91.82	89.06	90.42	82.52
CASP	**95.96**	91.05	93.44	87.69	**92.95**	88.85	90.85	83.24
STFFGA	95.13	92.19	93.64	88.04	92.04	89.57	90.79	83.13
PHENet	94.54	**92.80**	**93.66**	**88.08**	92.14	**89.78**	**90.94**	**83.39**

for operational deployment where weather conditions are unpredictable.

Precision–recall analysis reveals that most methods tend to favor precision over recall under foggy conditions, likely due to conservative detection strategies to avoid false positives from fog-induced artifacts. PHENet achieves a more balanced tradeoff, maintaining recall rates above 88% even in heavy fog while preserving high precision. This balance is crucial for practical applications where both missed detections and false alarms carry significant costs.

In addition, PHENet was evaluated on the original WHU-CD and LEVIR-CD datasets to verify its performance under normal weather conditions, as shown in Table IV. These clear-weather results serve as a no-fog baseline, enabling a direct quantitative comparison with light-, medium-, and heavy-fog scenarios. Despite being specifically designed for fog-degraded scenarios, PHENet demonstrates strong generalization capability on clean images, achieving the highest IoU of 88.08% on WHU-CD and 83.39% on LEVIR-CD, along with the best $F1$-scores of 93.66% and 90.94%, respectively. These results confirm that PHENet does not compromise performance on clear-weather imagery while providing substantial improvements under foggy conditions.

2) Visualization Results: Figs. 6 and 7 provide thr visual evidence of PHENet's superior detection quality across fog densities. In light fog conditions, PHENet produces cleaner boundaries with fewer scattered false positives compared with existing methods, particularly evident in building edge regions. As fog density increases, performance gaps become more pronounced. Many existing methods produce fragmented detections or miss smaller structures, showing vulnerability to fog-induced contrast reduction. PHENet consistently preserves detection completeness and boundary precision through height-guided feature enhancement, effectively disambiguating fog artifacts from genuine changes. Heavy fog scenarios reveal critical limitations in current approaches, with methods producing either severely fragmented results or missing numerous changes. The physics-aware design of PHENet maintains detection quality even under severe degradation, correctly identifying changes that other methods fail to detect.

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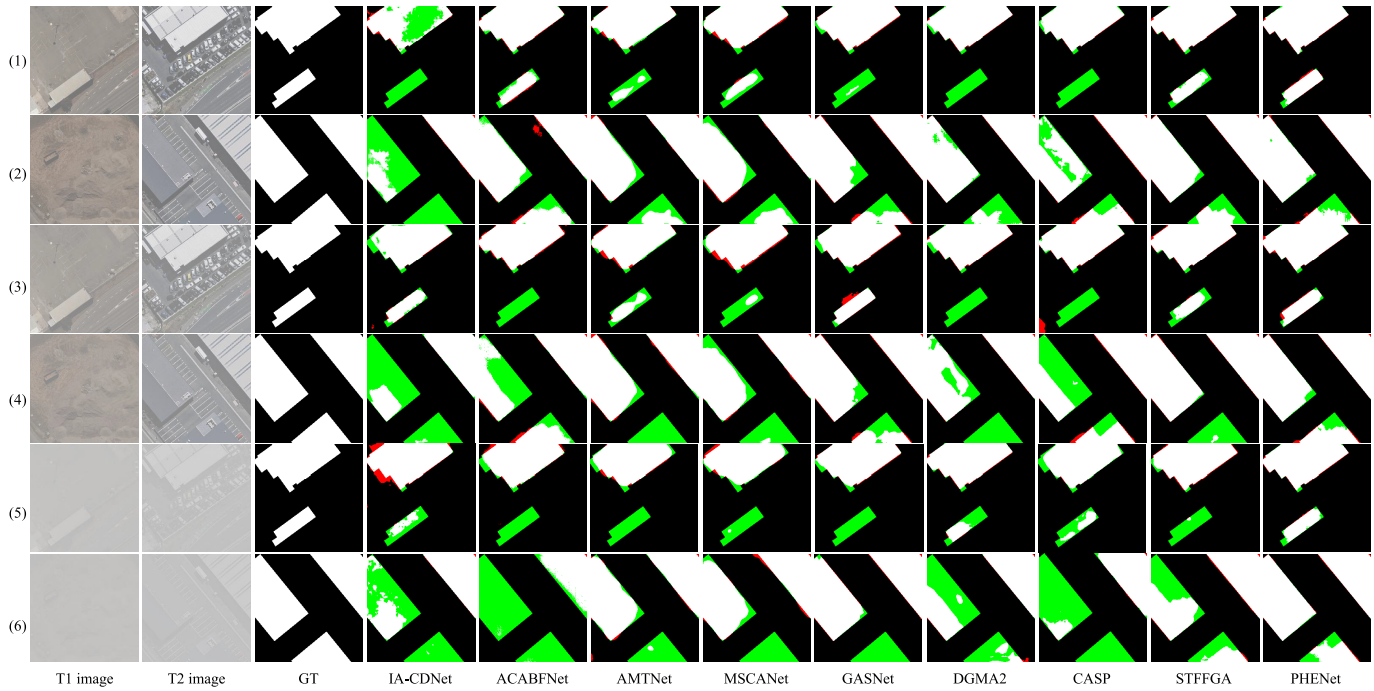


Fig. 6. Visual comparison of change detection results on WHU-CD dataset across fog densities. Rows (1) and (2), (3) and (4), and (5) and (6) show samples under light, medium, and heavy fog, respectively.

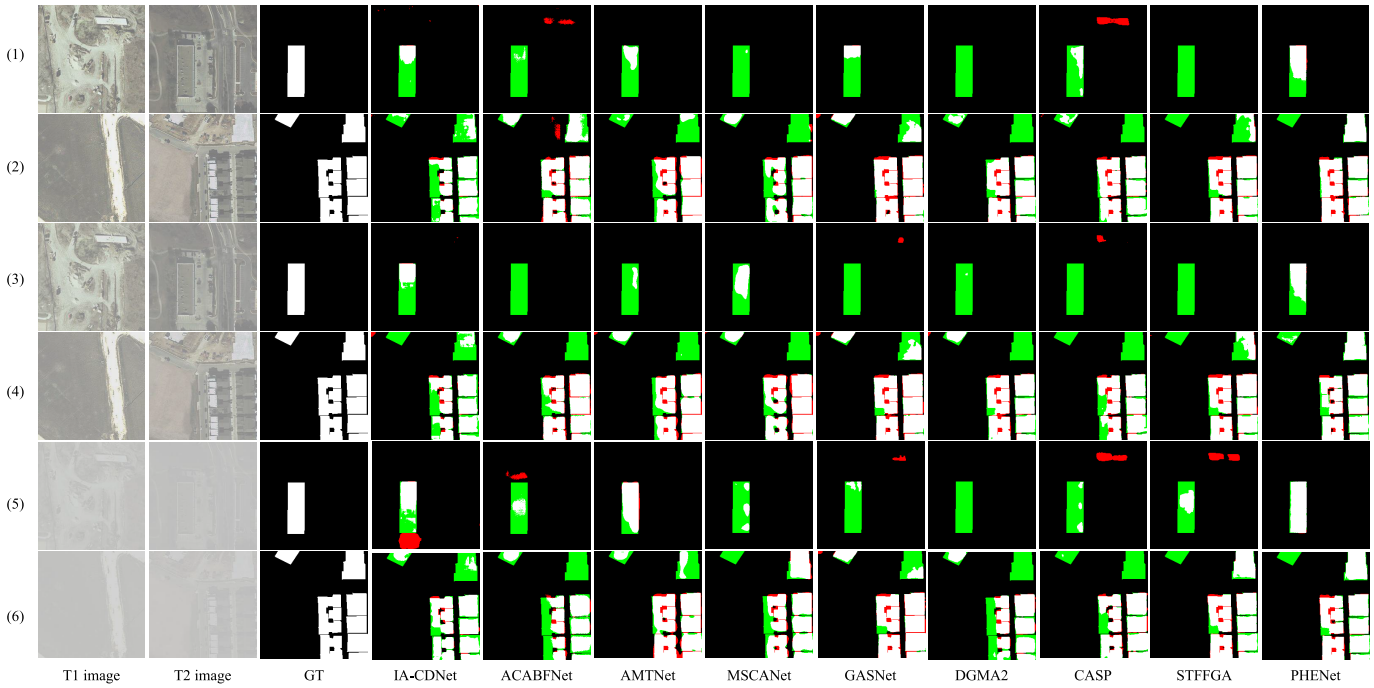


Fig. 7. Visual comparison of change detection results on LEVIR-CD dataset across fog densities. Rows (1) and (2), (3) and (4), and (5) and (6) show samples under light, medium, and heavy fog, respectively.

The visual analysis demonstrates that PHENet enables decisive classification with sharp boundaries across all fog conditions, validating the effectiveness of a physics-guided framework for robust change detection under foggy conditions.

C. Ablation Study

Comprehensive ablation experiments are performed to evaluate the contribution of individual components and critical architectural design choices.

1) Ablation Study for the Proposed Modules: Table V reports the ablation results under different fog levels on multiple datasets, where w/o denotes "without" and w/ denotes "with," respectively, and G refers to the adversarial training network, as illustrated in Fig. 5(b). The results show that HAMF and HED consistently improve performance across datasets and fog conditions. In contrast, PGARM alone (w/o G) yields limited gains and may even degrade performance in some cases, indicating that physical guidance without

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TABLE V
ABLATION STUDY OF THE PROPOSED MODULES

Dataset	HAMF	HED	PGARM (w/o G)	PGARM (w/ G)	light-foggy				medium-foggy				heavy-foggy			
					P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑
WHU-CD					94.12	91.32	92.70	86.39	92.91	92.00	92.45	85.96	93.22	91.21	92.20	85.54
	✓				94.07	92.24	93.15	87.17	94.13	91.90	93.00	86.92	94.19	90.71	92.42	85.91
	✓	✓			94.85	91.91	93.35	87.54	94.01	**92.68**	93.34	87.51	93.60	**92.58**	93.09	87.07
	✓	✓	✓		94.04	92.14	93.08	87.06	93.43	92.27	92.85	86.65	94.02	91.79	92.89	86.73
	✓	✓		✓	**95.02**	**92.65**	**93.82**	**88.36**	**95.41**	91.81	**93.57**	**87.92**	**94.97**	91.85	**93.38**	**87.58**
LEVIR-CD					93.27	87.60	90.35	82.39	**93.62**	87.26	90.33	82.36	**92.77**	86.81	89.69	81.31
	✓				**93.56**	87.65	90.51	82.66	92.13	**88.69**	90.38	82.45	91.52	88.38	89.92	81.69
	✓	✓			92.31	88.95	90.60	82.82	93.20	88.12	90.58	82.79	92.15	88.00	90.03	81.86
	✓	✓	✓		91.76	89.74	90.74	83.05	92.89	88.47	90.63	82.86	91.13	**88.79**	89.95	81.73
	✓	✓		✓	91.88	**89.86**	**90.86**	**83.25**	93.00	88.56	**90.73**	**83.03**	92.16	88.29	**90.19**	**82.13**

TABLE VI
ABLATION STUDY OF LOSS WEIGHT COEFFICIENTS

λ_1	λ_2	λ_3	P ↑	R ↑	F1 ↑	IoU ↑
1	1	1	93.66	92.10	92.87	86.69
1	1	0.5	94.36	91.68	93.00	86.92
1	0.5	0.5	94.56	91.83	93.17	87.22
0.5	0.5	0.5	94.36	92.33	93.33	87.50
0.2	0.5	0.5	**95.41**	91.81	**93.57**	**87.92**
0.2	0.6	0.6	94.99	92.02	93.48	87.76
0.2	0.4	0.4	95.09	91.79	93.41	87.64
0.2	0.2	0.2	93.51	**92.73**	93.12	87.12

adversarial supervision is insufficient. When PGARM is jointly optimized with the adversarial network G, significant and consistent improvements are achieved, and the full model delivers the best performance in all settings, validating the effectiveness and robustness of the proposed design.

The results validate our hypothesis that integrating physics-guided principles with data-driven learning yields superior fog robustness. The progressive improvements across different configurations demonstrate that each component addresses complementary aspects of the fog degradation problem: spectral artifact suppression through HAMF, robust change representation via height-enhanced difference computation, and physically plausible feature refinement through adversarial learning. The synergistic effect of all three modules underscores the effectiveness of our integrated approach to addressing foggy conditions in change detection.

2) Ablation Study for Loss Coefficients: The optimization of loss weight coefficients for the PAC loss significantly impacts model performance, as demonstrated in Table VI. This ablation is conducted on a representative foggy change detection setting to analyze the sensitivity of the model to different loss weight configurations. Initial experiments with equal weights ($\lambda_1 = \lambda_2 = \lambda_3 = 1$) yield 86.69% IoU, indicating that balanced contributions from all loss components provide a reasonable baseline. However, this configuration fails to account for the relative importance of different physical constraints.

Systematic exploration reveals that the optimal configuration $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.5$ achieves 87.92% IoU,

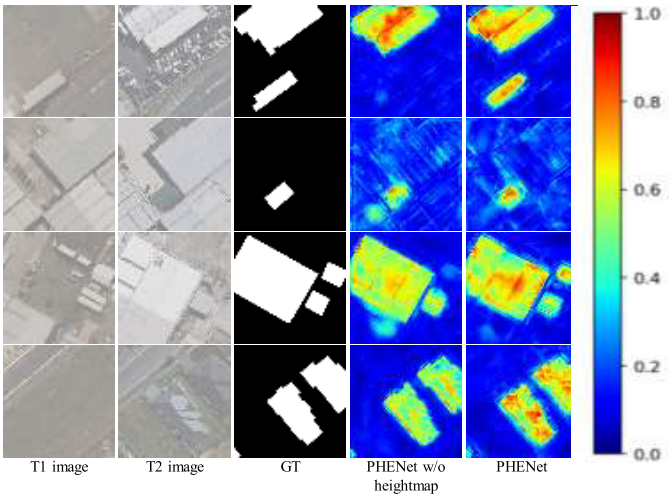


Fig. 8. Visualization of height information impact.

reflecting the inherent characteristics of foggy remote sensing imagery: while spatial smoothness of transmission maps remains important, excessive regularization can over-smooth fine building structures. The moderate weight on dark channel loss ensures physically plausible clear image recovery without dominating the optimization process.

3) Ablation Study for Heightmap: The contribution of height information is evaluated through comparative experiments presented in Table VII and visualized in Fig. 8. The absence of height information results in consistent performance degradation under varying fog conditions, indicating that elevation cues play a critical role in enhancing the robustness and discriminability of the proposed framework.

Fig. 8 provides visual evidence via decoder feature heatmaps extracted from the penultimate layer in the change detection decoder, revealing how height integration affects the network's internal representations. Without height information, feature activations exhibit less precise localization, with activation patterns that either miss actual changes or spread beyond the true change boundaries. In contrast, the PHENet produces more concentrated activations that better align with ground-truth change regions, particularly evident

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TABLE VII
ABLATION STUDY OF HEIGHT INFORMATION

Dataset	Models	light-foggy				medium-foggy				heavy-foggy			
		P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑	P↑	R↑	F1↑	IoU↑
WHU-CD	PHENet	**95.02**	**92.65**	**93.82**	**88.36**	**95.41**	**91.81**	**93.57**	**87.92**	**94.97**	**91.85**	**93.38**	**87.58**
	PHENet w/o heightmap	94.83	92.31	93.55	87.89	95.04	90.93	92.94	86.80	93.34	90.84	92.07	85.31
LEVIR-CD	PHENet	91.88	**89.86**	**90.86**	**83.25**	**93.00**	**88.56**	**90.73**	**83.03**	**92.16**	88.29	**90.19**	**82.13**
	PHENet w/o heightmap	**92.21**	88.61	90.37	82.44	92.68	88.51	90.55	82.72	91.53	**88.48**	89.98	81.78

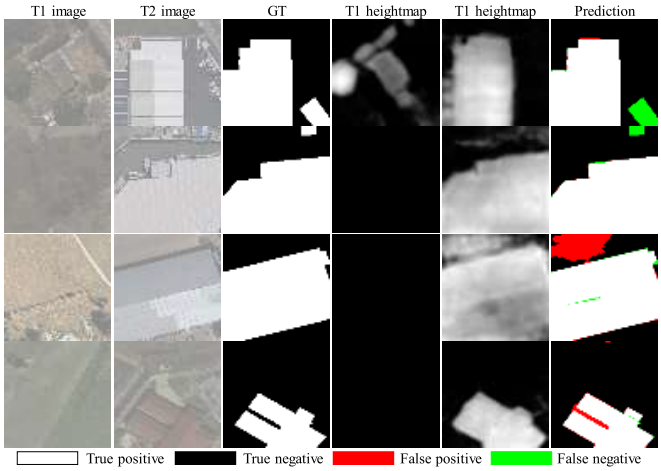


Fig. 9. Limitation analysis showing impact of low-quality height estimation on change detection.

in the improved boundary delineation and reduced false activations in unchanged areas. This comparison demonstrates that height-guided enhancement enables the network to more accurately distinguish genuine building changes from background variations, yielding cleaner and more confident predictions.

D. Discussion

However, accurate height estimation is critical for optimal performance, as illustrated in Fig. 9. Fig. 9 presents the representative failure cases stemming from height estimation errors. In the first two rows, the network fails to detect existing buildings in T1 and T2, respectively, resulting in false negatives. The third row shows the opposite scenario: flat terrain in T2 is erroneously assigned elevation values, causing false positives. The fourth row demonstrates a more subtle issue, where narrow gaps between adjacent buildings are not captured in the heightmap, leading to incorrect change classification. These examples highlight our method's sensitivity to both omission and commission errors in height estimation. This limitation underscores the need for change detection datasets that incorporate elevation annotations alongside 2-D change labels, enabling direct training with accurate elevation information and potentially improving performance under foggy conditions.

V. CONCLUSION

This article presents PHENet, a physics-guided height-enhanced network for RSCD under foggy conditions. By exploiting the physical relationship between atmospheric scattering and terrain elevation, our method introduces three key components: HAMF combining frequency-domain decomposition with spatial enhancement, HED-augmented difference fusion for ROBUST change representation, and physics-guided adversarial refinement for physically plausible feature learning. Experimental results demonstrate superior performance across all fog densities on both datasets.

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